RCE-2 v6.11

Causal Law Discovery — From Correlation to Causation

v6.11 adds the CausalDiscoveryEngine: instead of learning correlational laws (features that co-occur with outcomes), the substrate discovers CAUSAL laws (features that DETERMINE outcomes) via interventional queries. This distinguishes causation from correlation — a confounder that perfectly correlates with the outcome but doesn't cause it is correctly excluded from causal laws, because intervening on it doesn't change the outcome.

Project	Omega-X v6.11
Document	RCE-2 v6.11 Causal Law Discovery Report
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Status	[MEASURED] [IMPLEMENTED] [TESTED]
Test Suite	149 tests (134 prior + 15 causal), all passing
Mechanism	Pearl's do-calculus + Average Causal Effect (ACE)
Key Result	Confounders excluded from causal laws (ACE ≈ 0)

1. Executive Summary

This report documents the v6.11 Causal Law Discovery layer. All prior versions (v6.5-v6.10) learned CORRELATIONAL laws: 'features X co-occur with outcome Y'. But correlation does not imply causation. A feature may correlate with the outcome without causing it — it may be a confounder (causes both), a mediator (caused by the real cause), or a collider (caused by both). Correlational laws fail under intervention: forcing the confounder to a value doesn't change the outcome.

The v6.11 CausalDiscoveryEngine performs INTERVENTIONAL queries to distinguish causal features from confounders. For each feature, it asks: 'If I FORCE this feature to a specific value (do(feature=value)), does the outcome change?' If yes, the feature is CAUSAL. If no, the feature is CORRELATIONAL. Causal laws use ONLY causal features, making them robust to confounders — they predict what happens under intervention, the gold standard for causal inference.

Key Results

Metric	Value	Notes
Causal feature ACE > 0.5	Verified	Intervening changes outcome
Confounder ACE ≈ 0	Verified	Perfect correlation but no causal effect
Causal laws use only causal features	Verified	Confounders excluded from mask
Causal laws verified via intervention	Verified	confidence = 1.0
Intervening on confounder doesn't change prediction	Verified	Causal law robust to confounder
Test suite	149 tests	15 new causal discovery tests, all passing

2. Causal Discovery Engine

The CausalDiscoveryEngine wraps the substrate and a ground truth computation function. For each feature, it runs interventional queries (do(feature=value)) and measures the Average Causal Effect (ACE). Features with |ACE| > threshold are classified as causal; others are correlational or irrelevant.

2.1 Pearl's Do-Calculus

```
Observational:    P(Y | X=x)  — probability of Y given we OBSERVE X=x
Interventional:   P(Y | do(X=x))  — probability of Y given we FORCE X=x

ACE(X→Y) = P(Y=1 | do(X=1)) - P(Y=1 | do(X=0))

A feature is causal iff |ACE| > ε (default ε = 0.1)
```

The key distinction is between OBSERVATION and INTERVENTION. Observing X=1 tells us X is 1, but X may be 1 because of a confounder. Forcing X=1 (do(X=1)) breaks the confounding pathway — if Y still changes, X is causal; if Y doesn't change, X was just correlated.

2.2 Confounder Example

Consider a SEG admission system where 'has_z3_witness' correlates perfectly with 'admissible' because both are caused by 'evidence_chain_complete'. A correlational law would say 'has_z3_witness → admissible'. But intervening on has_z3_witness (setting it True without the evidence chain) does NOT make the computation admissible. The causal law correctly excludes has_z3_witness and uses only evidence_chain_complete.

2.3 Causal Law Verification

Causal laws are verified via interventional queries: the engine forces the causal features to their law-specified values on random base features, and checks if the outcome is always the expected one. Only laws that pass this intervention test are kept. This gives causal laws confidence = 1.0 — they have been verified under intervention, not just observation.

3. Confounder Robustness

The key benefit of causal laws is confounder robustness. A correlational law includes confounders in its mask and makes wrong predictions when the confounder is intervened on. A causal law excludes confounders and predicts correctly under intervention.

3.1 Measured Example

In the self-test, a computation was defined where feature 0 (evidence_chain) causes both feature 1 (has_witness) and the outcome. Feature 1 is a confounder: it correlates perfectly with the outcome (both are True iff feature 0 is True) but does not cause it.

```
Feature 0 (evidence_chain): ACE = 1.000, role = causal
Feature 1 (has_witness):    ACE = 0.000, role = irrelevant
Feature 2 (risk_class):     ACE = 0.000, role = irrelevant

Causal law uses ONLY feature 0:
  law features = (0,)
  law values = (1.0,)
  outcome = True
  confidence = 1.0 (verified via intervention)
```

Despite feature 1 correlating perfectly with the outcome, the causal engine correctly identifies it as non-causal (ACE = 0) and excludes it from the causal law. The law predicts correctly even when feature 1 is intervened on (set to 0 while feature 0 is 1) — exactly the behavior we want.

4. Conclusion

v6.11 adds causal inference to the substrate. The CausalDiscoveryEngine distinguishes causal features from confounders via interventional queries, producing causal laws that are robust to confounding. This is the strongest form of law the substrate can learn: it predicts what happens under intervention, not just observation.

The substrate now has a complete learning hierarchy with causal foundations:

```
Level 0: States              (v6.5 — observed computations)
Level 1: Correlational Laws  (v6.5 — compress states into regions)
Level 2: Refined Laws        (v6.9 — self-correcting via verification)
Level 3: Transferred Laws    (v6.10 — transfer across types via isomorphism)
Level 4: Causal Laws         (v6.11 — robust to confounders via intervention)
Level 5: Meta-Causal         (future — learn which causal structures transfer)
```

Each level adds a stronger guarantee: correlational laws predict observation, refined laws predict verification, transferred laws predict across types, and causal laws predict intervention. The causal layer is the foundation — without it, the substrate cannot distinguish features that cause outcomes from features that merely co-occur with them.

What v6.11 Does NOT Claim

v6.11 does not claim the causal discovery is complete. The engine uses a simple ACE test (binary interventions on each feature independently). Real causal discovery may require: (1) conditional interventions (do(X=x | Z=z)), (2) structural causal model discovery (learning the full DAG), (3) handling unobserved confounders. These are candidates for v6.12. v6.11 also requires a ground truth compute_fn for interventional queries — in production, this is an offline training cost, not an online capability.

Appendix A. File Inventory

Artifact	Location	Notes
Causal discovery engine	rce/rce2_causal_discovery.py	ACE + interventional queries + causal laws
v6.11 tests	tests/test_rce2_causal_discovery.py	15 tests, all passing
This report	rce/RCE2_V611_REPORT.pdf	PDF

Appendix B. How to Reproduce

```
# Run all 149 tests
cd /home/z/my-project/download
python -m pytest tests/test_rce2.py tests/test_rce2_adversarial.py \
  tests/test_rce2_substrate.py tests/test_rce2_density_gate.py \
  tests/test_rce2_adaptive_gate.py tests/test_rce2_reuse_classifier.py \
  tests/test_rce2_law_refiner.py tests/test_rce2_meta_learner.py \
  tests/test_rce2_causal_discovery.py

# Quick self-test
python -c "from rce.rce2_causal_discovery import self_test; self_test()"
```